

Greenline (Forest / Recon, Concealment, Ambush Warfare) Map Design

Executive Summary

Greenline is a dense, hyper-realistic battlefield designed for concealment, stealth, ambush warfare, and sensor-limited engagements. The terrain features thick woodland, rolling hills, hidden trails, sensitive rivers, narrow stations, shallow streams, and dense undergrowth. The map emphasizes Obsidian Protocol's autonomous stealth behavior, sensor fusion under heavy occlusion, terrain-reliant networks, and tactics under forest constraints. In AR (Real Late/AR), the map anchors at ~4-5 m clearance distance, covers a 10x10 m physical area measured "1 km" of forest territory. Key features include ambush-heavy choke zones, concealed movement paths, elevated vantage points, and multi-route infiltration corridors.

Design Goals & Gameplay

Forest Recon & Concealment Combat

Greenline prioritizes stealth, scouting, and ambush tactics. Dense vegetation and uneven terrain create natural concealment and unpredictable sightlines.

Autonomy & Stealth Pathfinding

Autonomous AI selects hidden trails, avoids open clearings, and uses undergrowth for concealment. Units adapt to shifting foliage density and sensor occlusion.

Sensor & Information Warfare

Thick foliage, humidity, and terrain occlusion degrade sensors. Relay towers on ridges restore clarity. Destroying relays creates fog-of-war pockets across forest sectors.

Logistics & Resources

Logging sites, ranger stations, and forest depots serve as resupply nodes. Supply convoys must navigate narrow trails and avoid ambush-prone choke zones.

Deployment & Progress

A hard deployment budget (~10,000 points) forces players to choose between stealth infilt., recon drops, or light vehicles. Blue deploys from the southwest ridge station; Red deploys from the northeast logging ridge.

Spatial Constraints (Nreal AR)

Device Capabilities

Target laptop: "15" diagonal FHD, 1800x1080 per-eye resolution. Optimal viewing distance ~4-5 m.

Playable Area & Scaling

10x10 m physical area x 1 km² forest zone (1:100 scale). Designed for tabletop AR.

Tracking & Occlusion

10x10 m physical area x 1 km² forest zone (1:100 scale). Depth-mesh (if available) provides limited occlusion. Virtual objects render over real-world view; assume minimal real-world occlusion.

Layout Overview

Coordinate System

1000x1000 grid (1 Unit = 1 m). Origin (0,0) = southwest corner.

Key Locations

Blue HQ (Ranger Station Command) — (0,0). 140x140 m Forest outpost with recon vehicles and drone path.

Red HQ (Logging Ridge Base) — (1000,1000). 100x100 m Elevated ridge with heavy machinery and cannon tower.

Central Forest Base — (500,450). 200x250 m Dense woodland, undergrowth, fallen trees, low visibility.

Logging Depot — (100,800). 100x100 m Fuel tanker, machinery, fuel drums.

Old Cabin Cluster — (800,100). 120x120 m Abandoned cabins, interior routes, close-range combat.

Stream Network — (0,450) to (400,900) Shallow water corridors with ford points.

Hidden Trail Route — (300,0) to (400,300) Narrow concealed path ideal for stealth infiltration.

Relay Station A — (450,150). 100 m Low-range sensor station point.

Relay Station B — (110,850). 100 m Cannon ridge tower and recon hub.

Neutral Forest Outpost — (600,900) Dense woods zone.

Chokepoints & Sightlines

Hidden Trail Route

Perfect for ambushes, extremely narrow and concealed.

Stream Crossings

Shallow water slows movement, vulnerable during crossing.

Cabin Cluster

Interior building routes, close-range engagements.

Forest Basin

Heavy foliage, unpredictable sightlines and stealth combat.

Sightlines

Relay Station A provides long-range sensor coverage (~200 m). Dense foliage creates fog-of-war pockets.

Deployment Zones

Blue Deployment: NW corner ridge + fuel ridge south.

Red Deployment: NE corner ridge + elevated artillery.

Neutral Forest Base: Outpost at (600,900) for drone drops.

Assets & Environment

Terrain

Forest hills, undergrowth, streams, ridges, fallen logs.

Vegetation

Dense trees, shrubs, ferns, grass patches, moss.

Buildings & Props

Military: Watchtowers, antennas, bunkers. Civilian: Cabins, sheds, logging machinery. Logistics: Trucks, crates, fuel drums, repair drones.

Effects

Fog, humidity haze, falling leaves, dust, insects.

Audio

Wind through trees, birds, insects, distant machinery, radio chatter.

LOD & Budget

Asset Cate. Count

Forest Tree 1 250k 250k

Underbrush 4 80k 22k

Buildings 15 10k 75k

Road/Trails 4 10k 12k

Vegetation 200 2k 400k

Props 50 10k 20k

Total — — 7.6M

Texture Budget

2048 for terrain/buildings, 1024 for props, 512x1/5 compression. Target 100 MB.

Unreal / Nreal Integration

Engine & Plugins

Unreal AR Template + Nreal SDK (NREAL). ARKit/ARCore enabled.

Rendering

Forward rendering, stationary lights, baked GI, aggressive culling.

Anchors

Single spatial anchor for entire map.

Interaction

Phone gesture for pan/zoom; HUD as screen-space widgets.

Multitasker

Client-server architecture, offload AI/physics to server.

Implementation Plan & Testing

Milestones

1. Prototype Layout — Month 1 Block-out forest basin, ridges, cabins, AR anchor test.

2. Core Systems — Month 2 Basic AI, stealth pathfinding, autonomous behaviors.

3. AR Integration — Month 3 AR Anchors, scene detection, basic interactivity.

4. Content Fill — Month 4-5 Final models, textures, lighting, audio.

5. Gameplay Polish — Months 6-7 Obstacle, balance, VFX (fog, leaves, humidity haze).

6. Testing & Opt. — Month 8 Multitasker stress test, sensor validation.

Testing Plan

Autonomy

Verify stealth logic, hidden trail navigation, ambush behavior.

Comms/Sensors

Test relay detection, fog-of-war updates, LOS checks.

Performance

400 Hz AR render on target device.

QA Checklist

Navigation, concealment behavior, deployment budget enforcement, anchor drift, sensor accuracy.